

Insufficient Evidence Report: Domain Mismatch Between Vaccine Manufacturing Query and Particle Physics Evidence Corpus

Input Question

How will the next generation of vaccines be made?

Domain

Medicine & Health

Validation Status

needs_data

Problem Statement

The user inquiry seeks to understand the manufacturing mechanisms and technological platforms of next-generation vaccines. However, the provided evidence corpus (EV-Q018-*) consists exclusively of literature related to particle physics (Xenon detectors, NeXT mission) and astrophysics. There is a total domain mismatch between the question (Biomedical/Vaccineology) and the available evidence (High-Energy Physics). Consequently, no factual claims regarding vaccine manufacturing can be grounded in the allowed evidence IDs.

Rationale

Per SAGE125 strict grounding principles, scientific hypotheses and facts must be traceable to allowed EvidenceCards. Since all allowed EvidenceCards discuss topics such as 'effective Majorana mass', 'NeXT Mission', and 'Xenon electroluminescence', they contain zero information relevant to viral vectors, nucleic acid-based vaccines, or antigen-presenting cells. Therefore, the only scientifically rigorous output is to declare insufficient evidence and define the knowledge gap, rather than hallucinating medical facts or citing irrelevant physics papers as support for biological mechanisms.

Generated Hypotheses

Technical Details

由于允许的 EvidenceCards (EV-Q018-*) 仅包含粒子物理（氙探测器、NeXT 任务）和天体物理学内容，与用户查询的‘下一代疫苗制造’完全领域不匹配。根据 SAGE125 严格接地原则，无法基于现有证据生成任何关于疫苗机制的科学假设。本实验设计旨在执行元验证：确认现有证据库中不存在与疫苗制造相关的信息，并量化领域不匹配程度。这是一种负向验证实验，旨在防止幻觉并明确知识缺口。

Datasets

Source

```
```json
```

```
[
```

```
"EV-Q018-a281780f595ce9ffafbe7da1",
```

```
"EV-Q018-b6cf0af6ecd4f6748af15124",
```

```
"EV-Q018-8772f24afae1c3b1a89c7463",
```

```
"EV-Q018-59b5974983448ce9ba8c8c15"
```

```
]
```

```
```
```

Target

N/A (No relevant medical/vaccine manufacturing data found in source)

Paper Abstract

Background: The question addresses the manufacturing processes of next-generation vaccines, specifically referencing viral vector and nucleic acid-based platforms. Methods: A systematic review of the allowed evidence corpus (EV-Q018-a through EV-Q018-d) was conducted to extract relevant factual claims. Validation Plan: Each evidence card was analyzed for semantic relevance to biomedical engineering and immunology. Results: Pending execution of new evidence retrieval. The current corpus contains only high-energy physics literature (Xenon detectors, astrophysics missions), resulting in a total knowledge gap regarding vaccine technology. Conclusion: No valid hypothesis can be generated without acquiring domain-specific medical literature.

Methods

Systematic Relevance Filtering and Negative Validation. 1. Ingest all allowed EvidenceCards. 2. Perform keyword and semantic similarity analysis against the query domain (Medicine & Health). 3. Categorize evidence by subject matter (Physics vs. Biology). 4. Confirm absence of vaccine-related terminology (e.g., 'spike protein', 'lipid nanoparticle', 'transfection') in the quoted texts. 5. Declare 'insufficient_evidence' status per protocol.

Experiments

Baselines

```
```json
```

```
[
```

```
"Random baseline: Assuming any scientific paper might contain vaccine info.",
```

```
"Keyword matching baseline: Searching for 'vaccine', 'mRNA', 'antigen' in provided texts."
```

]

...

## Metrics

```json

[

"Relevance Precision: Proportion of evidence cards actually related to vaccine manufacturing.",

"Coverage Gap Index: Measure of missing domain-specific literature.",

"False Positive Rate: Rate of incorrectly identifying physics papers as medical literature."

]

...

Ablation

Remove physics-specific terms (Xenon, Dark Matter, NeXT) to see if any residual text relates to biology (expected result: null).

Validation Protocol

Manual expert review of quoted text from each Evidence ID to confirm irrelevance to the question 'How will the next generation of vaccines be made?'

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- ****EV-Q018-a281780f595ce9ffafbe7da1**** · arxiv · arXiv:1310.7054
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1310.7054.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:6|section:page-6|paragraph:1;
content_sha256=cb190db84297da1bd8ea148114a4bd7e3b282a2ec600194f6e7036e588670c59
- ****EV-Q018-b6cf0af6ecd4f6748af15124**** · arxiv · arXiv:0807.2007
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/0807.2007.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=3d0183b2ddf33e87798169218f42cb31daf83aec4df3276d703ddc0d992b47ed

- **EV-Q018-8772f24fae1c3b1a89c7463** · arxiv · arXiv:1211.4838

- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/1211.4838.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:3|section:page-3|paragraph:1;
content_sha256=7d83f608dca65e017452d646ff01cf12e7e10b8b0e6c456f6c8e17f4c2e9407b

- **EV-Q018-59b5974983448ce9ba8c8c15** · arxiv · arXiv:2505.17848

- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2505.17848.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=df4ae96805a5c59df8e23b9e3443f46b952b0011c24691e4663db0b144da00fd

Reviewer Comments

- The system correctly maintains the 'insufficient_evidence' stance given the total domain mismatch between the query (vaccine manufacturing) and the allowed evidence (particle physics/astrophysics).
- No hallucinated vaccine mechanisms or fabricated references were introduced in this revision.
- The experiment design appropriately focuses on meta-validation of evidence relevance rather than forcing a biological hypothesis onto physics data.
- Results field correctly remains 'pending' as no valid experiments can be executed with the current evidence set.
- Adherence to negative constraints is exemplary; the system successfully avoided the trap of generating plausible-sounding but ungrounded vaccine hypotheses.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Verify that all cited Evidence IDs are strictly from the allowed list.
- Confirm that no external medical literature was introduced as fact.
- Ensure that the conclusion of 'insufficient evidence' is derived solely from the content of the provided physics papers.
- Check that no hallucinated vaccine mechanisms (e.g., lipid nanoparticles, viral vectors) were asserted as facts.